

Glove-based Orientation and Articulation Tracker

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Chapter 1

AIS2104 Industrielle styrsystemer Project

This repository contains the code and configuration for the AIS2104 Industrielle styrsystemer project. The project is built using [PlatformIO](#) and the Arduino framework for the ESP32 (Arduino Nano ESP32 board). This project takes use of the 9-DoF [IMU](#) (Inertial Measurement Unit) to track the orientation of the hand, and a way of creating a "Virtual Joystick" to control manipulate a 6-DoF Robot Arm. It uses flex sensors to toggle tracking of the virtual joystick and orientation, and publishes it via [MQTT](#) to a broker.

1.1 Table of Contents

- Overview
- Installation
- System Architecture

1.2 Overview

This project is for controlling A UR5e robot via [MQTT](#), I have provided a detailed description on how to initialize this project!

1.3 Installation

1.3.1 Prerequisites

- **PlatformIO:** Install PlatformIO IDE or the PlatformIO Core.
 - For the CLion, follow the instructions [here](#).

1.3.2 Clone the Repository

1. Open your terminal.
2. Clone the repository by running:

```
git clone git@github.com:HermanGran/Master-of-Machines.git
```

1.3.3 PlatformIO setup

To compile and build the project, run the following:

```
pio init --ide clion
```

Then run

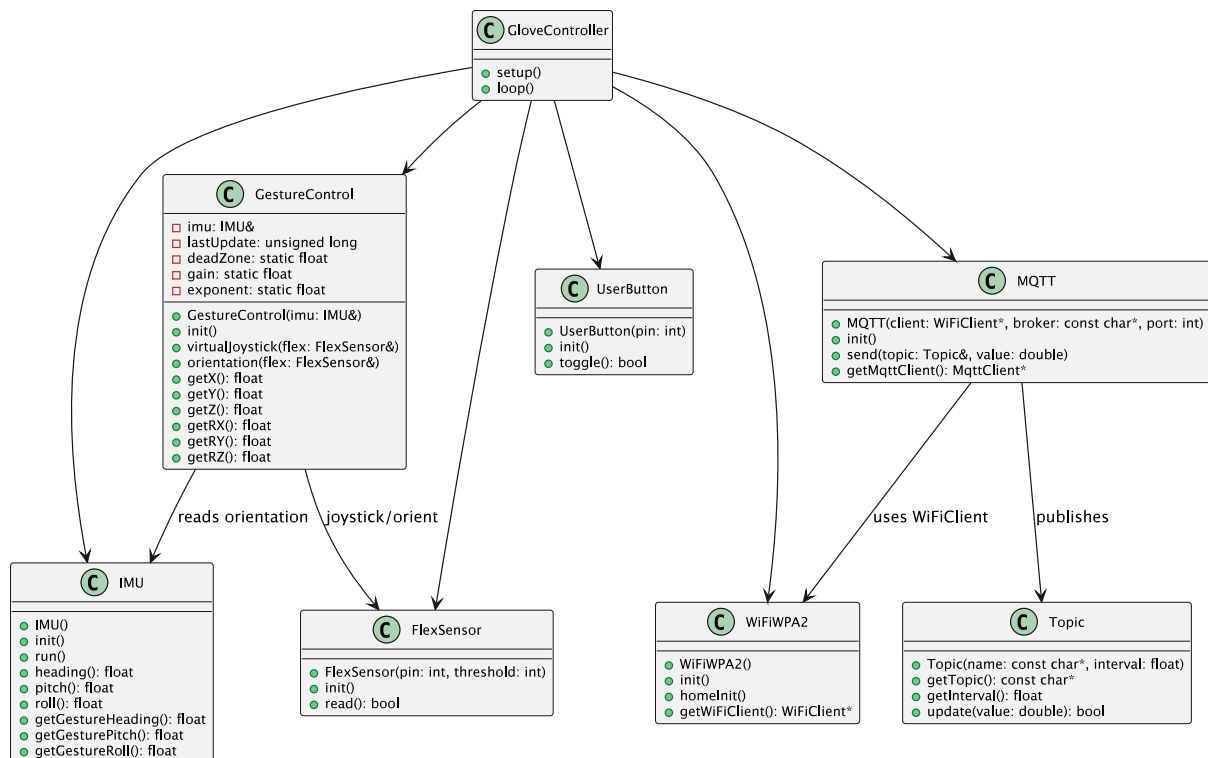
```
pio run
```

To open the serial monitor, run:

```
pio device monitor
```

1.4 System Architecture Diagram (UML)

The figure below describes the system architecture of the glove, generated by plantuml package. Throughout the code is also doxygen commenting style used, to generate a doxygen documentation.



Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

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Chapter 4

Class Documentation

4.1 FlexSensor Class Reference

```
#include <FlexSensor.hpp>
```

Collaboration diagram for FlexSensor:

FlexSensor
- inputPin
- threshold
+ FlexSensor()
+ init()
+ read()

Public Member Functions

- [FlexSensor](#) (int inputPin_, int threshold_)
- void [init](#) ()
- bool [read](#) ()

Private Attributes

- int [inputPin](#)
- int [threshold](#)

4.1.1 Detailed Description



FlexSensor

□
inputPin: int

□
threshold: int

●
FlexSensor(int, int)

●
init()

●
read(): bool

Class for flex sensors, reading and returning if activated above a certain threshold. Can also use for speed (according to how much you flex the sensor).

4.1.2 Constructor & Destructor Documentation

4.1.2.1 FlexSensor()

```
FlexSensor::FlexSensor (
    int inputPin_,
    int threshold_)
```

Initializing instance of the class

Parameters

<i>inputPin</i> ↔ —	Defining input pin
<i>threshold</i> ↔ —	Define threshold value for bool

4.1.3 Member Function Documentation

4.1.3.1 init()

```
void FlexSensor::init ()
```

Initializing the flex sensor, used in setup

4.1.3.2 read()

```
bool FlexSensor::read ()
```

Returning true or false for flex above threshold

Returns

4.1.4 Member Data Documentation

4.1.4.1 inputPin

```
int FlexSensor::inputPin [private]
```

4.1.4.2 threshold

```
int FlexSensor::threshold [private]
```

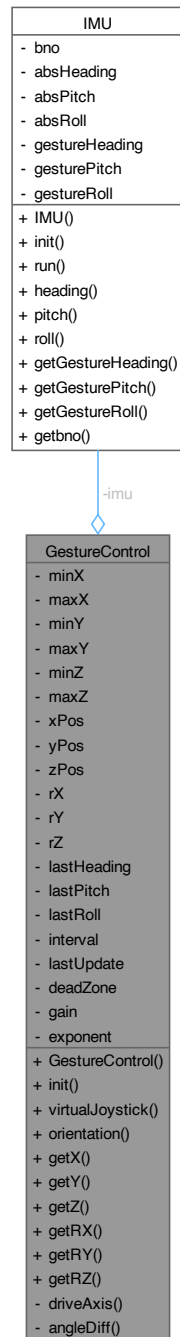
The documentation for this class was generated from the following files:

- include/Connections/[FlexSensor.hpp](#)
- src/Connections/[FlexSensor.cpp](#)

4.2 GestureControl Class Reference

```
#include <GestureControl.hpp>
```

Collaboration diagram for GestureControl:



Public Member Functions

- [GestureControl](#) ([IMU](#) &imu_)

- void `init` ()
- void `virtualJoystick` (`FlexSensor` &`flexSensor`)
- void `orientation` (`FlexSensor` &`flexSensor`)
- float `getX` () const
- float `getY` () const
- float `getZ` () const
- float `getRX` () const
- float `getRY` () const
- float `getRZ` () const

Private Member Functions

- void `driveAxis` (float `deltaDeg`, float &`pos`)
- float `angleDiff` (float `a`, float `b`)

Private Attributes

- `IMU` & `imu`
- float `minX`
- float `maxX`
- float `minY`
- float `maxY`
- float `minZ`
- float `maxZ`
- float `xPos` = 300
- float `yPos` = 50
- float `zPos` = 100
- float `rX` = 3.14
- float `rY` = 0
- float `rZ` = 1.83
- float `lastHeading` = 0
- float `lastPitch` = 0
- float `lastRoll` = 0
- float `interval`
- unsigned long `lastUpdate` = 0

Static Private Attributes

- static constexpr float `deadZone` = 10.0f
- static constexpr float `gain` = 0.001f
- static constexpr float `exponent` = 1.5f

4.2.1 Detailed Description

Class for gesture control sensor, like a virtual joystick

4.2.2 Constructor & Destructor Documentation

4.2.2.1 GestureControl()

```
GestureControl::GestureControl (
    IMU & imu_)
```

Constructor

Parameters

<i>imu</i> ↔	An instance of the imu class
—	

4.2.3 Member Function Documentation

4.2.3.1 angleDiff()

```
float GestureControl::angleDiff (
    float a,
    float b) [private]
```

4.2.3.2 driveAxis()

```
void GestureControl::driveAxis (
    float deltaDeg,
    float & pos) [private]
```

4.2.3.3 getRX()

```
float GestureControl::getRX () const
```

Getter function

Returns

Rotation along x axis

4.2.3.4 getRY()

```
float GestureControl::getRY () const
```

Getter function

Returns

Rotation along y axis

4.2.3.5 getRZ()

```
float GestureControl::getRZ () const
```

Getter function

Returns

Rotation along z axis

4.2.3.6 getX()

```
float GestureControl::getX () const
```

Getter function

Returns

X position

4.2.3.7 getY()

```
float GestureControl::getY () const
```

Getter function

Returns

Y position

4.2.3.8 getZ()

```
float GestureControl::getZ () const
```

Getter function

Returns

Z pos

4.2.3.9 init()

```
void GestureControl::init ()
```

Init function, used in setup

4.2.3.10 orientation()

```
void GestureControl::orientation (  
    FlexSensor & flexSensor)
```

Function for updating total orientation, not relative, but absolute orientation.

Parameters

<i>flexSensor</i>	An instance of flex sensor class, used for toggling
-------------------	---

4.2.3.11 virtualJoystick()

```
void GestureControl::virtualJoystick (  
    FlexSensor & flexSensor)
```

Virtual joystick function for tracking relative position from start of flex sensor

Parameters

<i>flexSensor</i>	An instance of flex sensor class, used for toggling of tracking
-------------------	---

4.2.4 Member Data Documentation

4.2.4.1 deadZone

```
float GestureControl::deadZone = 10.0f [static], [constexpr], [private]
```

4.2.4.2 exponent

```
float GestureControl::exponent = 1.5f [static], [constexpr], [private]
```

4.2.4.3 gain

```
float GestureControl::gain = 0.001f [static], [constexpr], [private]
```

4.2.4.4 imu

```
IMU& GestureControl::imu [private]
```

4.2.4.5 interval

```
float GestureControl::interval [private]
```

4.2.4.6 lastHeading

```
float GestureControl::lastHeading = 0 [private]
```

4.2.4.7 lastPitch

```
float GestureControl::lastPitch = 0 [private]
```

4.2.4.8 lastRoll

```
float GestureControl::lastRoll = 0 [private]
```

4.2.4.9 lastUpdate

```
unsigned long GestureControl::lastUpdate = 0 [private]
```

4.2.4.10 maxX

```
float GestureControl::maxX [private]
```

4.2.4.11 maxY

```
float GestureControl::maxY [private]
```

4.2.4.12 maxZ

```
float GestureControl::maxZ [private]
```

4.2.4.13 minX

```
float GestureControl::minX [private]
```

4.2.4.14 minY

```
float GestureControl::minY [private]
```

4.2.4.15 minZ

```
float GestureControl::minZ [private]
```

4.2.4.16 rX

```
float GestureControl::rX = 3.14 [private]
```

4.2.4.17 rY

```
float GestureControl::rY = 0 [private]
```

4.2.4.18 rZ

```
float GestureControl::rZ = 1.83 [private]
```

4.2.4.19 xPos

```
float GestureControl::xPos = 300 [private]
```

4.2.4.20 yPos

```
float GestureControl::yPos = 50 [private]
```

4.2.4.21 zPos

```
float GestureControl::zPos = 100 [private]
```

The documentation for this class was generated from the following files:

- [include/Control/GestureControl.hpp](#)
- [src/Control/GestureControl.cpp](#)

4.3 IMU Class Reference

```
#include <IMU.hpp>
```

Collaboration diagram for IMU:

IMU
- bno - absHeading - absPitch - absRoll - gestureHeading - gesturePitch - gestureRoll
+ IMU() + init() + run() + heading() + pitch() + roll() + getGestureHeading() + getGesturePitch() + getGestureRoll() + getbno()

Public Member Functions

- [IMU](#) ()
- void [init](#) ()
- void [run](#) ()
- float [heading](#) ()
- float [pitch](#) ()
- float [roll](#) ()
- float [getGestureHeading](#) ()
- float [getGesturePitch](#) ()
- float [getGestureRoll](#) ()
- Adafruit_BNO055 & [getbno](#) ()

Private Attributes

- Adafruit_BNO055 [bno](#)
- float [absHeading](#)
- float [absPitch](#)
- float [absRoll](#)
- float [gestureHeading](#)
- float [gesturePitch](#)
- float [gestureRoll](#)

4.3.1 Detailed Description

 IMU
❑ bno: Adafruit_BNO055 ❑ absHeading: float ❑ absPitch: float ❑ absRoll: float ❑ gestureHeading: float ❑ gesturePitch: float ❑ gestureRoll: float
● IMU() ● init() ● run() ● heading(): float ● pitch(): float ● roll(): float ● getGestureHeading(): float ● getGesturePitch(): float ● getGestureRoll(): float ● getbno(): Adafruit_BNO055&

Class for initializing the BNO055 [IMU](#) sensor, also retrieving sensor data

4.3.2 Constructor & Destructor Documentation

4.3.2.1 IMU()

```
IMU::IMU ()
```

Constructor for the class, creating an instance of the bno sensor using the Adafruit package

4.3.3 Member Function Documentation

4.3.3.1 getbno()

```
Adafruit_BNO055 & IMU::getbno ()
```

Getter function for bno instance from the adafruit package

Returns

4.3.3.2 getGestureHeading()

```
float IMU::getGestureHeading ()
```

Getter function for specific heading used in gesture control class

Returns

4.3.3.3 getGesturePitch()

```
float IMU::getGesturePitch ()
```

Getter function for specific pitch used in gesture control class

Returns

4.3.3.4 getGestureRoll()

```
float IMU::getGestureRoll ()
```

Getter function for specific roll used in gesture control class

Returns

4.3.3.5 heading()

```
float IMU::heading ()
```

Getter function for heading

Returns

4.3.3.6 init()

```
void IMU::init ()
```

Function for initializing the class, used in setup

4.3.3.7 pitch()

```
float IMU::pitch ()
```

Getter function for Pitch

Returns

4.3.3.8 roll()

```
float IMU::roll ()
```

Getter function for Roll

Returns

4.3.3.9 run()

```
void IMU::run ()
```

Function for running sensor, calculating quaternions

4.3.4 Member Data Documentation

4.3.4.1 absHeading

```
float IMU::absHeading [private]
```

4.3.4.2 absPitch

```
float IMU::absPitch [private]
```

4.3.4.3 absRoll

```
float IMU::absRoll [private]
```

4.3.4.4 bno

```
Adafruit_BNO055 IMU::bno [private]
```

4.3.4.5 gestureHeading

```
float IMU::gestureHeading [private]
```

4.3.4.6 gesturePitch

```
float IMU::gesturePitch [private]
```

4.3.4.7 gestureRoll

```
float IMU::gestureRoll [private]
```

The documentation for this class was generated from the following files:

- [include/Connections/IMU.hpp](#)
- [src/Connections/IMU.cpp](#)

4.4 MQTT Class Reference

```
#include <MQTT.hpp>
```

Collaboration diagram for MQTT:

MQTT
- mqttClient - broker_ - port_ - lastMillis - lastValue
+ MQTT() + init() + send() + getMqttClient()

Public Member Functions

- [MQTT](#) (WiFiClient *wifiClient, const char *broker, int port)
- void [init](#) ()
- void [send](#) (Topic &topic, double value)
- MqttClient * [getMqttClient](#) ()

Private Attributes

- MqttClient * [mqttClient](#)
- const char * [broker_](#)
- int [port_](#)
- unsigned long [lastMillis](#)
- double [lastValue](#)

4.4.1 Detailed Description

 MQTT
❑ mqttClient: MqttClient* ❑ broker_: const char* ❑ port_: int ❑ lastMillis: unsigned long ❑ lastValue: double
● MQTT(WiFiClient* wifiClient, const char* broker, int port) ● init() ● send(topic: Topic&, value: double) ● getMqttClient(): MqttClient*

Class for creating [MQTT](#) Client for publishing

4.4.2 Constructor & Destructor Documentation**4.4.2.1 MQTT()**

```
MQTT::MQTT (
    WiFiClient * wifiClient,
    const char * broker,
    int port)
```

Constructor for [MQTT](#) class

Parameters

<i>wiFiClient</i>	Needs an instance of wifiClient to work
<i>broker</i>	Broker IP address
<i>port</i>	Broker port

4.4.3 Member Function Documentation**4.4.3.1 getMqttClient()**

```
MqttClient * MQTT::getMqttClient ()
```

Getter function for pointer to mqttclient

Returns**4.4.3.2 init()**

```
void MQTT::init ()
```

Function for initializing the class, used in void setup

4.4.3.3 send()

```
void MQTT::send (
    Topic & topic,
    double value)
```

Function for publishing to broker

Parameters

<i>topic</i>	Topic name
<i>value</i>	Value to send

4.4.4 Member Data Documentation**4.4.4.1 broker_**

```
const char* MQTT::broker_ [private]
```

4.4.4.2 lastMillis

```
unsigned long MQTT::lastMillis [private]
```

4.4.4.3 lastValue

```
double MQTT::lastValue [private]
```

4.4.4.4 mqttClient

```
MqttClient* MQTT::mqttClient [private]
```

4.4.4.5 port_

```
int MQTT::port_ [private]
```

The documentation for this class was generated from the following files:

- include/Connections/[MQTT.hpp](#)
- src/Connections/[MQTT.cpp](#)

4.5 Topic Class Reference

```
#include <Topic.hpp>
```

Collaboration diagram for Topic:

Topic
- lastMillis - interval - topic - lastValue
+ Topic() + getLastMillis() + setLastMillis() + getTopic() + getInterval() + updateLastValue() + update()

Public Member Functions

- [Topic](#) (const char *[topic](#), float [interval](#))
- unsigned long [getLastMillis](#) ()
- void [setLastMillis](#) (unsigned long [lastMillis](#))
- const char * [getTopic](#) ()
- float [getInterval](#) ()
- void [updateLastValue](#) (double value)
- bool [update](#) (double value)

Private Attributes

- unsigned long [lastMillis](#)
- float [interval](#)
- const char * [topic](#)
- double [lastValue](#)

4.5.1 Detailed Description

 Topic
❑ lastMillis: unsigned long ❑ interval: float ❑ topic: const char* ❑ lastValue: double
● Topic(topic: const char*, interval: float) ● getLastMillis(): unsigned long ● setLastMillis(lastMillis: unsigned long) ● getTopic(): const char* ● getInterval(): float ● updateLastValue(value: double) ● update(value: double): bool

[Topic](#) class for creating topics to be published

4.5.2 Constructor & Destructor Documentation

4.5.2.1 Topic()

```
Topic::Topic (  
    const char * topic,  
    float interval)
```

Constructor for class

Parameters

<i>topic</i>	Topic you want to publish
<i>interval</i>	Interval to publish the topic

4.5.3 Member Function Documentation

4.5.3.1 getInterval()

```
float Topic::getInterval ()
```

Getter function for interval

Returns

Returns Interval

4.5.3.2 getLastMillis()

```
unsigned long Topic::getLastMillis ()
```

Getter function for last millis

Returns

Returning lastMillis

4.5.3.3 getTopic()

```
const char * Topic::getTopic ()
```

Getter function for topic

Returns

Returning Topic

4.5.3.4 setLastMillis()

```
void Topic::setLastMillis (  
    unsigned long lastMillis)
```

Setter function for last millis

Parameters

<i>lastMillis</i>	The millis you want to assign
-------------------	-------------------------------

4.5.3.5 update()

```
bool Topic::update (  
    double value)
```

Used for checking of the last value sent is the same, in that case, not sending value to broker

Parameters

<i>value</i>	value to send
--------------	---------------

Returns

returning boolean if last value is different, returns true

4.5.3.6 updateLastValue()

```
void Topic::updateLastValue (  
    double value)
```

Used for updating last value sent

Parameters

<i>value</i>	last value sent
--------------	-----------------

4.5.4 Member Data Documentation**4.5.4.1 interval**

```
float Topic::interval [private]
```

4.5.4.2 lastMillis

```
unsigned long Topic::lastMillis [private]
```

4.5.4.3 lastValue

```
double Topic::lastValue [private]
```

4.5.4.4 topic

```
const char* Topic::topic [private]
```

The documentation for this class was generated from the following files:

- include/Connections/[Topic.hpp](#)
- src/Connections/[Topic.cpp](#)

4.6 UserButton Class Reference

```
#include <UserButton.hpp>
```

Collaboration diagram for UserButton:

UserButton
- pin - debounce - lastPressed - toggled - risingEdgeBool - risingEdgeBool0
+ UserButton() + init() + toggle()

Public Member Functions

- [UserButton](#) (int pin_)
- void [init](#) ()
- bool [toggle](#) ()

Private Attributes

- int [pin](#)
- double [debounce](#) = 500
- unsigned long [lastPressed](#) = 0
- bool [toggled](#) = false
- bool [risingEdgeBool](#)
- bool [risingEdgeBool0](#)

4.6.1 Detailed Description

 C UserButton

- ❑ `pin: int`
- ❑ `debounce: double = 500`
- ❑ `lastPressed: unsigned long = 0`
- ❑ `toggled: bool = false`
- ❑ `risingEdgeBool: bool`
- ❑ `risingEdgeBool0: bool`

- `UserButton(pin_: int)`
- `init()`
- `toggle(): bool`

Class for initializing pushbutton and some function for reading

4.6.2 Constructor & Destructor Documentation

4.6.2.1 UserButton()

```
UserButton::UserButton (
    int pin_)
```

Constructor function, internal pull up

Parameters

<i>pin</i> ↔	input pin number
—	

4.6.3 Member Function Documentation

4.6.3.1 init()

```
void UserButton::init ()
```

Init function, used in setup

4.6.3.2 toggle()

```
bool UserButton::toggle ()
```

Toggle function when input pin pulled low

Returns

returning true or false in a toggle fashion

4.6.4 Member Data Documentation

4.6.4.1 debounce

```
double UserButton::debounce = 500 [private]
```

4.6.4.2 lastPressed

```
unsigned long UserButton::lastPressed = 0 [private]
```

4.6.4.3 pin

```
int UserButton::pin [private]
```

4.6.4.4 risingEdgeBool

```
bool UserButton::risingEdgeBool [private]
```

4.6.4.5 risingEdgeBool0

```
bool UserButton::risingEdgeBool0 [private]
```

4.6.4.6 toggled

```
bool UserButton::toggled = false [private]
```

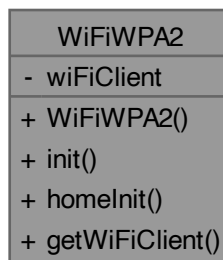
The documentation for this class was generated from the following files:

- [include/Connections/UserButton.hpp](#)
- [src/Connections/UserButton.cpp](#)

4.7 WiFiWPA2 Class Reference

```
#include <WiFiWPA2.hpp>
```

Collaboration diagram for WiFiWPA2:



Public Member Functions

- [WiFiWPA2](#) ()
- void [init](#) ()
- void [homeInit](#) ()
- WiFiClient * [getWiFiClient](#) ()

Private Attributes

- WiFiClient * [WiFiClient](#)

4.7.1 Detailed Description

Class for connecting the ESP32 to wifi via WPA2 connection Needed for school ID verification

4.7.2 Constructor & Destructor Documentation

4.7.2.1 WiFiWPA2()

```
WiFiWPA2::WiFiWPA2 ()
```

Constructor for this class

4.7.3 Member Function Documentation

4.7.3.1 getWiFiClient()

```
WiFiClient * WiFiWPA2::getWiFiClient ()
```

Getter function for pointer to wificlient

Returns

pointer to wificlient

4.7.3.2 homeInit()

```
void WiFiWPA2::homeInit ()
```

Added initialization for home wifi

4.7.3.3 init()

```
void WiFiWPA2::init ()
```

Initializing the class, used in void setup

4.7.4 Member Data Documentation

4.7.4.1 WiFiClient

```
WiFiClient* WiFiWPA2::WiFiClient [private]
```

The documentation for this class was generated from the following files:

- [include/Connections/WiFiWPA2.hpp](#)
- [src/Connections/WiFiWPA2.cpp](#)

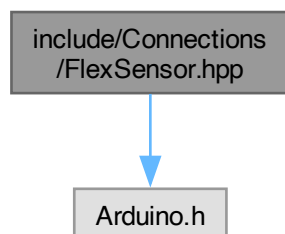
Chapter 5

File Documentation

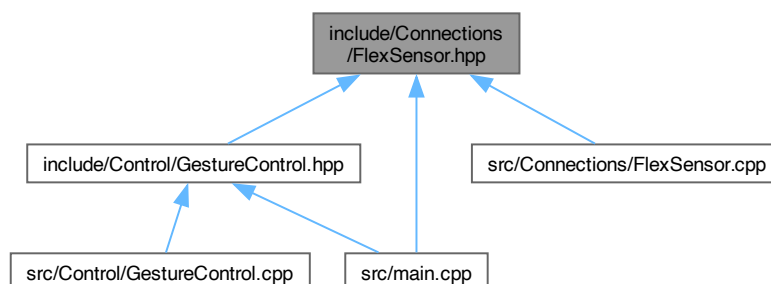
5.1 include/Connections/FlexSensor.hpp File Reference

```
#include "Arduino.h"
```

Include dependency graph for FlexSensor.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [FlexSensor](#)

5.2 FlexSensor.hpp

[Go to the documentation of this file.](#)

```

00001 //
00002 // Created by Herman Hårstad Gran on 29/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_FLEXSENSOR_HPP
00006 #define MOM_GLOVE_FLEXSENSOR_HPP
00007
00008 #include "Arduino.h"
00009
00021
00022
00027 class FlexSensor{
00028 public:
00034     FlexSensor(int inputPin_, int threshold_);
00035
00039     void init();
00040
00045     bool read();
00046
00047 private:
00048     int inputPin;
00049     int threshold;
00050 };
00051
00052 #endif //MOM_GLOVE_FLEXSENSOR_HPP

```

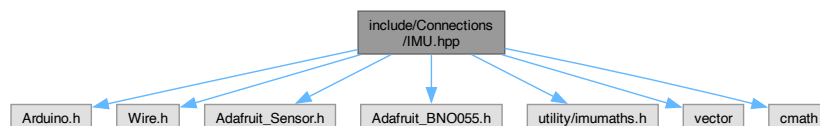
5.3 include/Connections/IMU.hpp File Reference

```

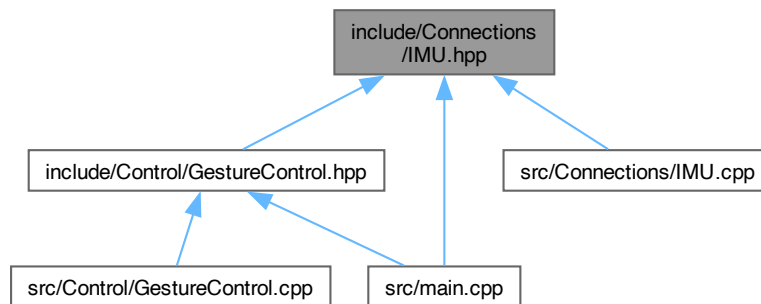
#include "Arduino.h"
#include "Wire.h"
#include "Adafruit_Sensor.h"
#include "Adafruit_BNO055.h"
#include "utility/imumaths.h"
#include <vector>
#include <cmath>

```

Include dependency graph for IMU.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `IMU`

5.4 IMU.hpp

[Go to the documentation of this file.](#)

```

00001 //
00002 // Created by Herman Hårstad Gran on 08/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_IMU_HPP
00006 #define MOM_GLOVE_IMU_HPP
00007
00008 #include "Arduino.h"
00009 #include "Wire.h"
00010 #include "Adafruit_Sensor.h"
00011 #include "Adafruit_BNO055.h"
00012 #include "utility/imumaths.h"
00013 #include <vector>
00014 #include <cmath>
00015
00039
00043 class IMU{
00044 public:
00048     IMU();
00049
00053     void init();
00054
00058     void run();
00059
00064     float heading();
00065
00070     float pitch();
00071
00076     float roll();
00077
00082     float getGestureHeading();
00083
00088     float getGesturePitch();
00089
00094     float getGestureRoll();
00095
00100     Adafruit_BNO055& getbno();
00101
00102 private:
00103     Adafruit_BNO055 bno;
00104
00105     float absHeading;
00106     float absPitch;

```

```

00107     float absRoll;
00108
00109     float gestureHeading;
00110     float gesturePitch;
00111     float gestureRoll;
00112 };
00113
00114 #endif //MOM_GLOVE_IMU_HPP

```

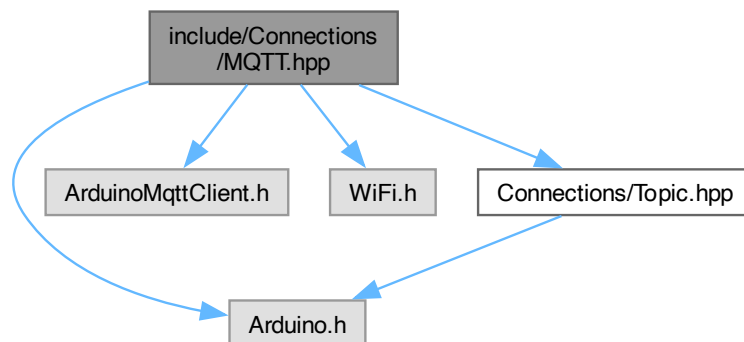
5.5 include/Connections/MQTT.hpp File Reference

```

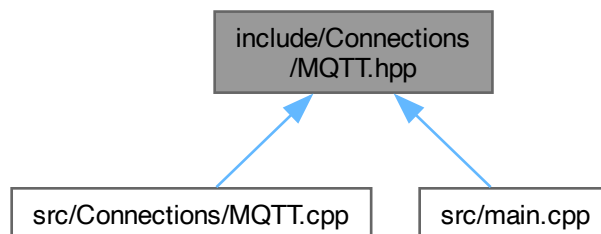
#include <Arduino.h>
#include <ArduinoMqttClient.h>
#include <WiFi.h>
#include "Connections/Topic.hpp"

```

Include dependency graph for MQTT.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `MQTT`

5.6 MQTT.hpp

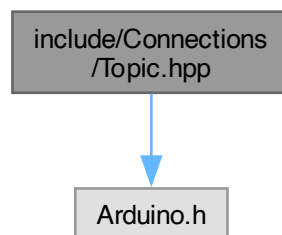
[Go to the documentation of this file.](#)

```
00001 //
00002 // Created by Herman Hårstad Gran on 02/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_MQTT_HPP
00006 #define MOM_GLOVE_MQTT_HPP
00007
00008 #include <Arduino.h>
00009 #include <ArduinoMqttClient.h>
00010 #include <WiFi.h>
00011 #include "Connections/Topic.hpp"
00012
00028
00032 class MQTT{
00033 public:
00040     MQTT(WiFiClient* wiFiClient, const char* broker, int port);
00041
00045     void init();
00046
00052     void send(Topic& topic, double value);
00053
00058     MqttClient* getMqttClient();
00059
00060 private:
00061     MqttClient* mqttClient;
00062     const char* broker_;
00063     int port_;
00064     unsigned long lastMillis;
00065     double lastValue;
00066 };
00067
00068 #endif //MOM_GLOVE_MQTT_HPP
```

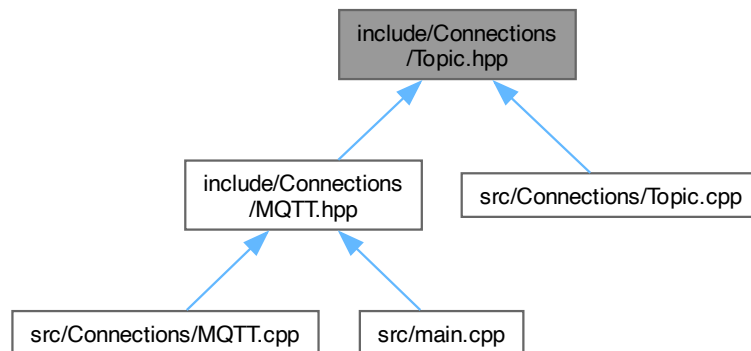
5.7 include/Connections/Topic.hpp File Reference

```
#include <Arduino.h>
```

Include dependency graph for Topic.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [Topic](#)

5.8 Topic.hpp

[Go to the documentation of this file.](#)

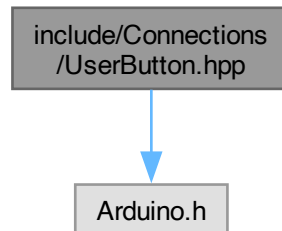
```

00001 //
00002 // Created by Herman Hårstad Gran on 03/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_TOPIC_HPP
00006 #define MOM_GLOVE_TOPIC_HPP
00007
00008 #include <Arduino.h>
00009
00010
00011 class Topic{
00012 public:
00013     Topic(const char* topic, float interval);
00014
00015     unsigned long getLastMillis();
00016
00017     void setLastMillis(unsigned long lastMillis);
00018
00019     const char* getTopic();
00020
00021     float getInterval();
00022
00023     void updateLastValue(double value);
00024
00025     bool update(double value);
00026
00027 private:
00028     unsigned long lastMillis;
00029     float interval;
00030     const char* topic;
00031     double lastValue;
00032 };
00033
00034 #endif //MOM_GLOVE_TOPIC_HPP
  
```

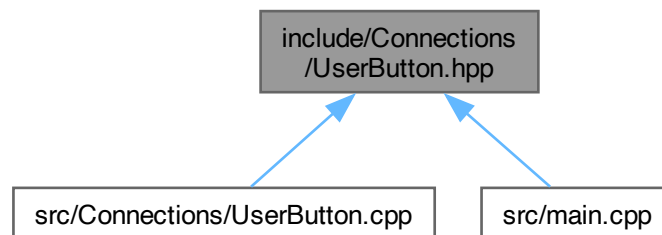
5.9 include/Connections/UserButton.hpp File Reference

```
#include "Arduino.h"
```

Include dependency graph for UserButton.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [UserButton](#)

5.10 UserButton.hpp

[Go to the documentation of this file.](#)

```
00001 //
00002 // Created by Herman Hårstad Gran on 08/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_USERBUTTON_HPP
00006 #define MOM_GLOVE_USERBUTTON_HPP
00007
00008 #include "Arduino.h"
00009
00025
```

```

00029 class UserButton{
00030 public:
00035     UserButton(int pin_);
00036
00040     void init();
00041
00046     bool toggle();
00047
00048 private:
00049     int pin;
00050     double debounce = 500;
00051     unsigned long lastPressed = 0;
00052     bool toggled = false;
00053     bool risingEdgeBool;
00054     bool risingEdgeBool0;
00055 };
00056
00057 #endif //MOM_GLOVE_USERBUTTON_HPP

```

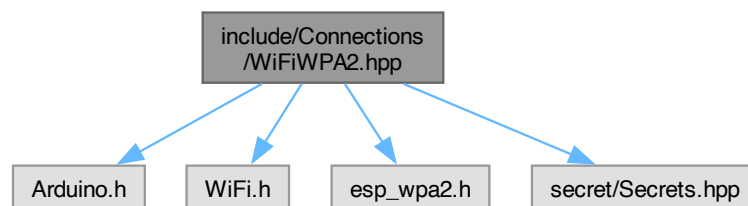
5.11 include/Connections/WiFiWPA2.hpp File Reference

```

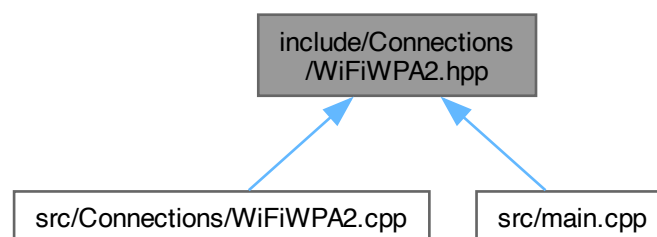
#include <Arduino.h>
#include <WiFi.h>
#include <esp_wpa2.h>
#include <secret/Secrets.hpp>

```

Include dependency graph for WiFiWPA2.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [WiFiWPA2](#)

5.12 WiFiWPA2.hpp

[Go to the documentation of this file.](#)

```

00001 //
00002 // Created by Herman Hårstad Gran on 02/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_WIFI_HPP
00006 #define MOM_GLOVE_WIFI_HPP
00007
00008 #include <Arduino.h>
00009 #include <WiFi.h>
00010 #include <esp_wpa2.h>
00011 #include <secret/Secrets.hpp>
00012
00017 class WiFiWPA2{
00018 public:
00022     WiFiWPA2();
00023
00027     void init();
00028
00032     void homeInit();
00033
00038     WiFiClient* getWiFiClient();
00039
00040 private:
00041     WiFiClient *wifiClient;
00042 };
00043
00044 #endif //MOM_GLOVE_WIFI_HPP

```

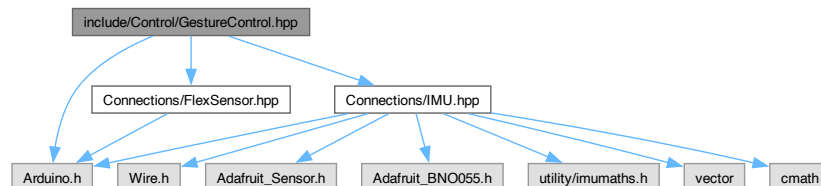
5.13 include/Control/GestureControl.hpp File Reference

```

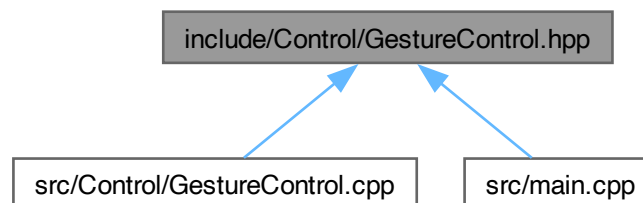
#include <Arduino.h>
#include "Connections/IMU.hpp"
#include <Connections/FlexSensor.hpp>

```

Include dependency graph for GestureControl.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [GestureControl](#)

5.14 GestureControl.hpp

[Go to the documentation of this file.](#)

```

00001 //
00002 // Created by Herman Hårstad Gran on 29/04/2025.
00003 //
00004
00005 #ifndef MOM_GLOVE_GESTERCONTROL_HPP
00006 #define MOM_GLOVE_GESTERCONTROL_HPP
00007
00008 #include <Arduino.h>
00009 #include "Connections/IMU.hpp"
00010 #include <Connections/FlexSensor.hpp>
00011
00012 class GestureControl{
00013 public:
00014     GestureControl(IMU& imu_);
00015
00016     void init();
00017
00018     void virtualJoystick(FlexSensor& flexSensor);
00019
00020     void orientation(FlexSensor& flexSensor);
00021
00022     float getX() const;
00023
00024     float getY() const;
00025
00026     float getZ() const;
00027
00028     float getRX() const;
00029
00030     float getRY() const;
00031
00032     float getRZ() const;
00033
00034 private:
00035     IMU& imu;
00036
00037     float minX, maxX, minY, maxY, minZ, maxZ;
00038
00039     float xPos = 300, yPos = 50, zPos = 100;
00040     float rX = 3.14, rY = 0, rZ = 1.83;
00041
00042     float lastHeading = 0, lastPitch = 0, lastRoll = 0;
00043
00044     float interval;
00045     unsigned long lastUpdate = 0;
00046
00047     // Updated variables (help from GPT)
00048     static constexpr float deadZone = 10.0f;
00049     static constexpr float gain = 0.001f; // How far one 'unit' moves the object, think this will be 1
00050     static constexpr float exponent = 1.5f;
00051
00052     // Function for movement
00053     void driveAxis(float deltaDeg, float& pos);
00054     float angleDiff(float a, float b);
00055 };
00056
00057 #endif //MOM_GLOVE_GESTERCONTROL_HPP

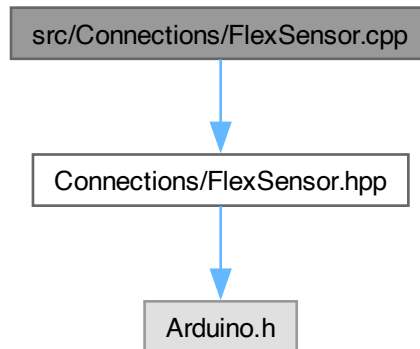
```

5.15 README.md File Reference

5.16 src/Connections/FlexSensor.cpp File Reference

```
#include "Connections/FlexSensor.hpp"
```

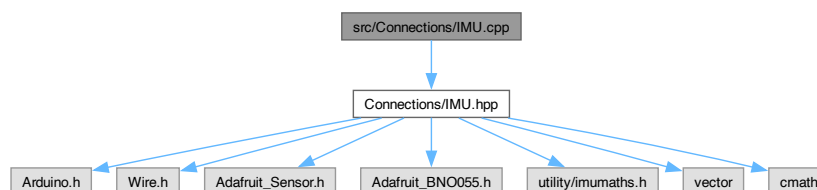
Include dependency graph for FlexSensor.cpp:



5.17 src/Connections/IMU.cpp File Reference

```
#include "Connections/IMU.hpp"
```

Include dependency graph for IMU.cpp:



Functions

- void [quaternionToEulerRad](#) (const imu::Quaternion &q, float &roll, float &pitch, float &yaw)

5.17.1 Function Documentation

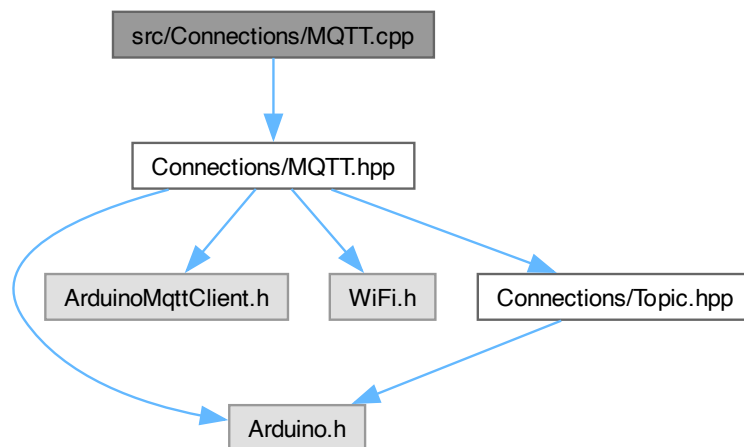
5.17.1.1 quaternionToEulerRad()

```
void quaternionToEulerRad (  
    const imu::Quaternion & q,  
    float & roll,  
    float & pitch,  
    float & yaw)
```

5.18 src/Connections/MQTT.cpp File Reference

```
#include "Connections/MQTT.hpp"
```

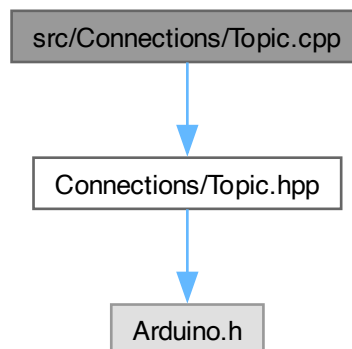
Include dependency graph for MQTT.cpp:



5.19 src/Connections/Topic.cpp File Reference

```
#include "Connections/Topic.hpp"
```

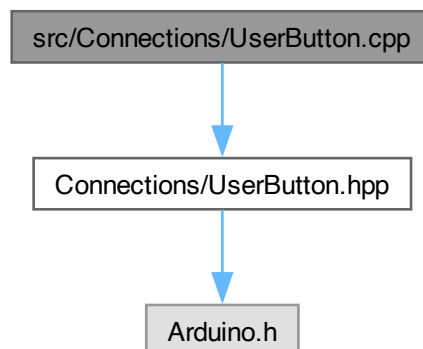
Include dependency graph for Topic.cpp:



5.20 src/Connections/UserButton.cpp File Reference

```
#include "Connections/UserButton.hpp"
```

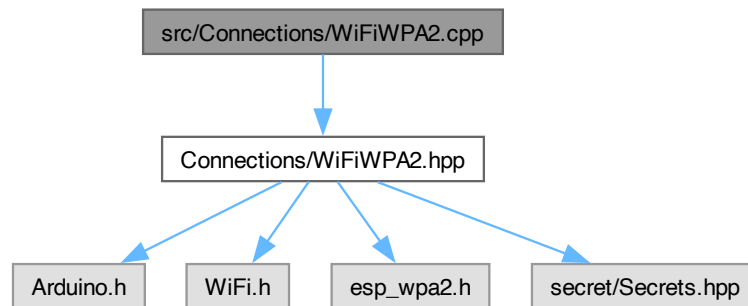
Include dependency graph for UserButton.cpp:



5.21 src/Connections/WiFiWPA2.cpp File Reference

```
#include "Connections/WiFiWPA2.hpp"
```

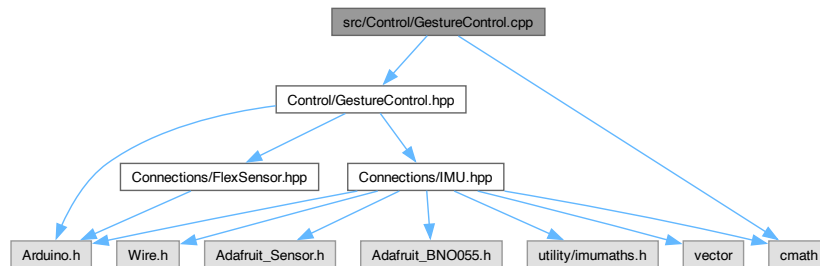
Include dependency graph for WiFiWPA2.cpp:



5.22 src/Control/GestureControl.cpp File Reference

```
#include "Control/GestureControl.hpp"
#include <cmath>
```

Include dependency graph for GestureControl.cpp:

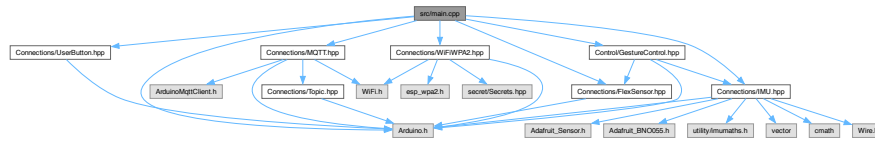


5.23 src/main.cpp File Reference

```
#include <Arduino.h>
#include <Connections/MQTT.hpp>
#include <Connections/WiFiWPA2.hpp>
#include <Connections/IMU.hpp>
#include <Connections/FlexSensor.hpp>
#include "Control/GestureControl.hpp"
```

```
#include "Connections/UserButton.hpp"
```

Include dependency graph for main.cpp:



Functions

- [MQTT mqtt](#) ([wifi.getClient\(\)](#), "129.241.30.177", 1883)
- void [setup](#) ()
- void [loop](#) ()

Variables

- [WiFiWPA2 wifi](#)
- [IMU bno_IMU](#)
- [Topic rxValue](#) ("mom/rxValue", 100)
- [Topic ryValue](#) ("mom/ryValue", 100)
- [Topic rzValue](#) ("mom/rzValue", 100)
- [Topic xValue](#) ("mom/xValue", 100)
- [Topic yValue](#) ("mom/yValue", 100)
- [Topic zValue](#) ("mom/zValue", 100)
- [FlexSensor toggleTrack](#) (4, 3350)
- [FlexSensor toggleOrientation](#) (5, 3350)
- [GestureControl gestureControl](#) ([bno_IMU](#))

5.23.1 Function Documentation

5.23.1.1 loop()

```
void loop ()
```

5.23.1.2 mqtt()

```
MQTT mqtt (
    wifi.  getClient(),
    "129.241.30.177" ,
    1883 )
```

5.23.1.3 setup()

```
void setup ()
```

5.23.2 Variable Documentation

5.23.2.1 bno_IMU

`IMU` `bno_IMU`

5.23.2.2 gestureControl

```
GestureControl gestureControl(bno_IMU) (  
    bno_IMU )
```

5.23.2.3 rxValue

```
Topic rxValue("mom/rxValue", 100) (  
    "mom/rxValue" ,  
    100 )
```

5.23.2.4 ryValue

```
Topic ryValue("mom/ryValue", 100) (  
    "mom/ryValue" ,  
    100 )
```

5.23.2.5 rzValue

```
Topic rzValue("mom/rzValue", 100) (  
    "mom/rzValue" ,  
    100 )
```

5.23.2.6 toggleOrientation

```
FlexSensor toggleOrientation(5, 3350) (  
    5 ,  
    3350 )
```

5.23.2.7 toggleTrack

```
FlexSensor toggleTrack(4, 3350) (  
    4 ,  
    3350 )
```

5.23.2.8 wifi

`WiFiWPA2` `wifi`

5.23.2.9 xValue

```
Topic xValue("mom/xValue", 100) (  
    "mom/xValue" ,  
    100 )
```

5.23.2.10 yValue

```
Topic yValue("mom/yValue", 100) (  
    "mom/yValue" ,  
    100 )
```

5.23.2.11 zValue

```
Topic zValue("mom/zValue", 100) (  
    "mom/zValue" ,  
    100 )
```


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